



SMSE: Social Media Stock Estimator

Short-Term Stock Movement Forecasting Using Reddit + Financial Data

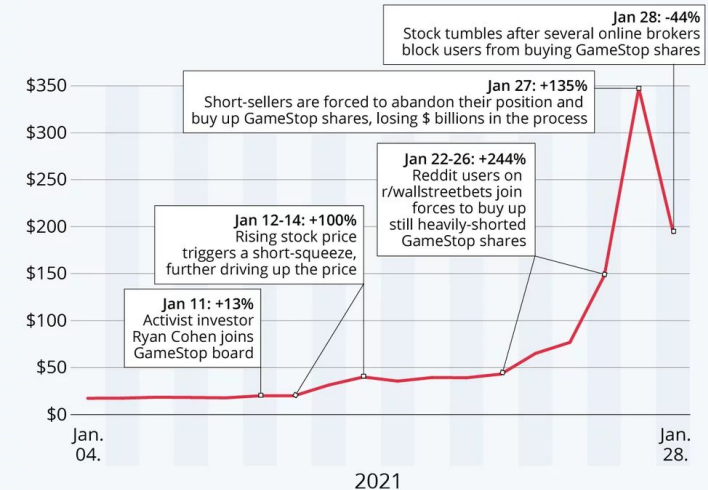
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Why Sentiment Matters in Modern Stock Market?

- Reddit-driven events like **GameStop** showed social media can dramatically affect stock prices.
- But sentiment signals online are **noisy and unstructured**.
- Investors lack tools that **convert Reddit discussions into measurable, predictive signals**.

Reddit vs. Wall Street: The GameStop Saga

Year-to-date stock performance
of GameStop (closing prices)



Source: Yahoo! Finance



statista

What are the earlier works?

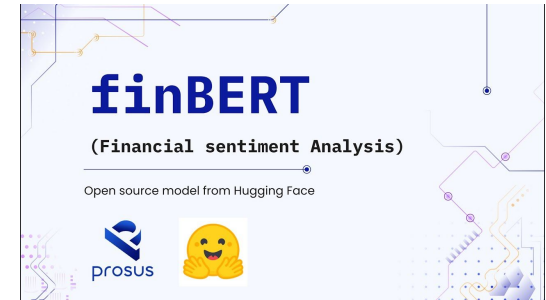
- Early work used **lexicon-based sentiment tools** (VADER, TextBlob)
While recent studies used **FinBERT** provide superior sentiment accuracy for investment discussions.
- Research highlights that **aggregate sentiment alone is weak**; conversation-level signals meaningfully improve forecasting power.
- Temporal modeling studies suggest **5–7 day sentiment effects**, but Reddit cycles are more volatile than news cycles, requiring flexible temporal features.

➔ Effective systems require **transformer sentiment, conversation-level features, temporal modeling, and engineered embeddings**

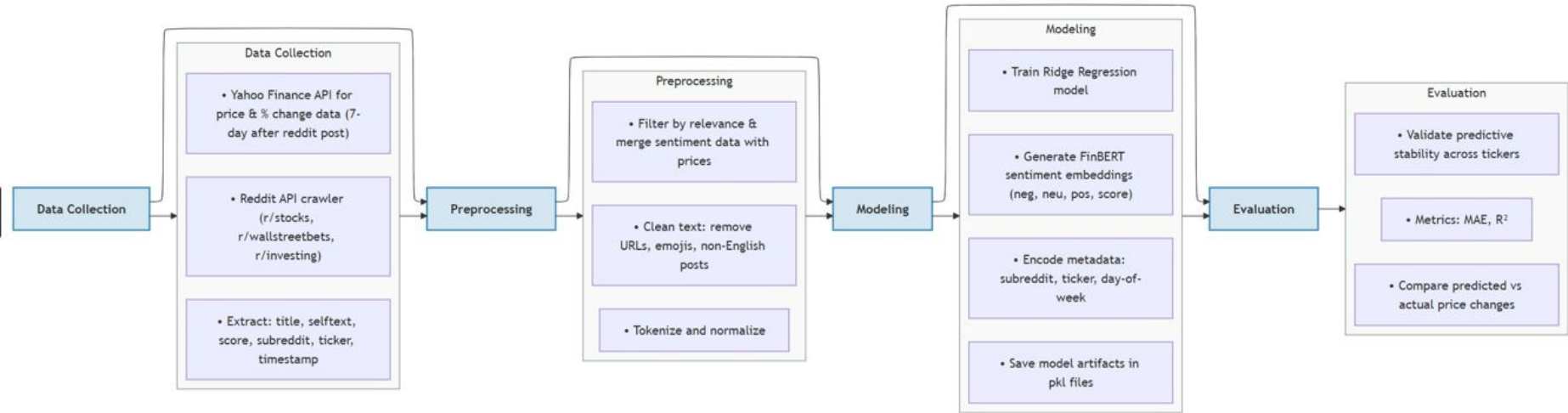
Our Solution & Goal

Transform unstructured Reddit discussions into structured, predictive insights about short-term stock movement.

- Data Ingestion Pipeline to create financial predicting model combining Reddit posts + sentiment analysis + multi-regression model.
- Interactive prototype for browsing sentiment trends and viewing model forecasts.



Overview Architecture of Data Ingestion Pipeline



Data Collection

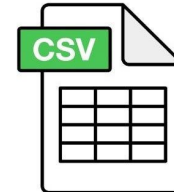
- PRAW scraper fetching posts for target large-cap tickers
- Filters: title + body mentions
- Saved fields:
 - Post text
 - Subreddit
 - Score / comments
 - Created timestamp
 - Ticker labels

Benefits

- Highly extensible to new tickers
- Full reproducibility for training / evaluation

TABLE I
REDDIT DATA SUMMARY (v2)

Metric	Value
Total Posts	5,206
Total Comments	80,141
Companies	18
Subreddits	4
Avg. Comments/Post	15.4
Data Version	v2



Pre-processing pipeline

Text Processing

- Lowercasing, URL removal, ticker normalization
- Stopword removal + emoji handling
- Bot detection + removal

FinBERT Output

- `neg_prob`, `neu_prob`, `pos_prob`
- Sentiment label (argmax)
- Continuous sentiment score = `pos - neg`

Yahoo Finance Ingestion

- Daily OHLCV for each ticker
- Align Reddit posts by **posting date**
- Compute:
 - **7-day return %**
 - 1-day / 3-day volatility
 - Market-open vs after-hours alignment

Modeling

- Extract **FinBERT sentiment** (neg/neu/pos) + **sentiment score**
- Generate **PCA-reduced embeddings** and add metadata (subreddit, ticker, time features)
- Train **ElasticNet regression** for 7-day return prediction

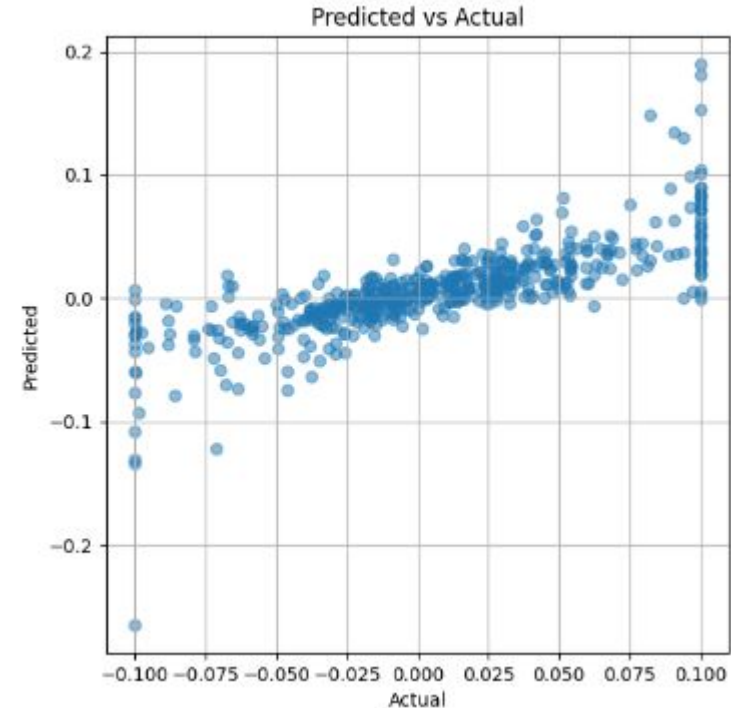
Evaluation

- Metrics: **MAE**, **R^2** , **Correlation**
- Validate performance across tickers and time periods
- Compare **predicted vs actual** 7-day returns

Results of Data Ingestion Pipeline

TABLE II
MODEL PERFORMANCE COMPARISON

Model	MAE↓	R^2 ↑	Corr↑
v3 ElasticNet + PCA(128)	0.0237	0.6206	0.7898
Phase-I RidgeCV (small dataset)	0.0205	0.3300	0.6900
v2 RidgeCV	0.0413	0.0159	0.1395
v2 LightGBM	0.0412	0.0072	0.1429
Baseline (mean)	0.045	0.0000	0.0000



User Survey Results

1. Social Media Influence Perception:

Most respondents rated influence **between 4-5**, acknowledging **significant impact**

2. Primary Concerns:

Manipulation and **misinformation**

Echo chambers and **herd mentality**

False positives and **noise in data**

Over Reliance on **sentiment over fundamentals**

3. Desired Features:

Transparency in data sources and methodology

Integration with traditional technical indicators

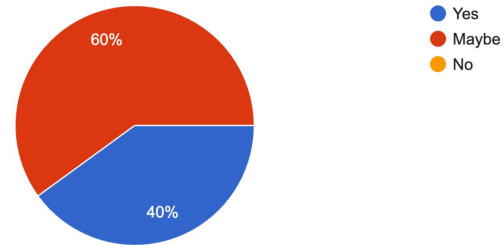
Community-level sentiment breakdown

Filtering by sentiment polarity and volume

Historical accuracy metrics for confidence calibration

Would you be interested in an app that also tracks your watchlist and summarizes social sentiment for your chosen stocks?

10 responses



User Interview Results

Interview insights from an active trader who had been given access to the initial prototype of the app:

- **Source visibility essential** for trust—users verify aggregated scores
- **Causality unclear**: Does sentiment lead or lag price?
- **Needs**: Tooltips, historical accuracy rates, drastic changes section
- **Key finding**: Transparency matters as much as accuracy

Figma Prototype Demo

Live demo link

- <https://www.figma.com/make/wOTjTkWq3yzsGb9ionRoAk/Social-Media-Stock-Estimator?node-id=0-1&p=f&t=VXfweCGsWCQoityw-0&fullscreen=1>



Conclusion and Future Work

The bottom line is that Reddit sentiment contains measurable signals for 7-day price movements.

- **What Works:** Simple features + ElasticNet ($R^2=0.62$, $r=0.79$) | Transparency > Accuracy
- **Limitations:** 3-month window, large-cap bias, small user sample
- **Future:** Longer datasets, multi-platform, real-time deployment, larger user studies
- **Key Takeaway:** Technical feasibility validated; trust requires interpretability



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Thank you!

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